familiar conceptions of space and time, while the astonishing truth of quantum reality lay carefully guarded by nature's sleights of hand.

In recent years, physicists have expended much effort in trying to explain nature's ruse—to figure out precisely how the fundamental laws of quantum physics morph into the classical laws that are so successful at explaining common experience—in essence, to figure out how the atomic and subatomic shed their magical weirdness when they combine to form macroscopic objects. Research continues, but much has already been learned. Let's look at some aspects of particular relevance to the question of time's arrow, but now from the standpoint of quantum mechanics.

Classical mechanics is based on equations that Newton discovered in the late 1600s. Electromagnetism is based on equations Maxwell discovered in the late 1800s. Special relativity is based on equations Einstein discovered in 1905, and general relativity is based on equations he discovered in 1915. What all these equations have in common, and what is central to the dilemma of time's arrow (as explained in the last chapter), is their completely symmetric treatment of past and future. Nowhere in any of these equations is there anything that distinguishes "forward" time from "backward" time. Past and future are on an equal footing.

Quantum mechanics is based on an equation that Erwin Schrödinger discovered in 1926.[6] You don't need to know anything about this equation beyond the fact that it takes as input the shape of a quantum mechanical probability wave at one moment of time, such as that in Figure 4.5, and allows one to determine what the probability wave looks like at any other time, earlier or later. If the probability wave is associated with a particle, such as an electron, you can use it to predict the probability that, at any specified time, an experiment will find the electron at any specified location. Like the classical laws of Newton, Maxwell, and Einstein, the quantum law of Schrödinger embraces an egalitarian treatment of time-future and time-past. A "movie" showing a probability wave starting like *this* and ending like *that* could be run in reverse—showing a probability wave starting like *that* and ending like *this*— and there would be no way to say that one evolution was right and the other wrong. Both would be equally valid solutions of Schrödinger's